

NdFeB Magnet Grades & GBD Premium Series

Standard (N-series) and Grain Boundary Diffusion enhanced grades with energy product (BH)max and temperature ratings

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Grain Boundary Diffusion (GBD) Technology Benefits

• Higher intrinsic coercivity (HcJ) with significantly less dysprosium/terbium usage

• Maintains remanence (Br) and energy product (BH)max while improving temperature stability

• Cost-effective premium grades for high-temperature applications

• Reduced rare earth element consumption aligns with environmental sustainability goals

Typical Applications by Grade Class

Grade Class	Temperature Range	Key Applications
Standard (N-series)	≤80°C	Consumer electronics, small DC motors, sensors, general industrial components
M-Class	100°C	Automotive sensors, oil/gas downhole tools, servo motors, industrial actuators
H-Class	120°C	Premium automotive motors, aerospace instruments, moderate-heat industrial equipment

SH-Class	150°C	Electric vehicle (EV) traction motors, spindle drives, high-performance industrial
UH/EH/AH	180-230°C	Advanced EV motors, industrial turbines, aerospace/defense, geothermal & downhole equipment
GBD Premium	180-230°C	Next-gen EV powertrains, high-efficiency wind turbines, medical imaging, extreme environments

All specifications based on industry-standard parameters. GBD technology offers enhanced performance and cost-effectiveness for demanding applications. Consult technical datasheets for specific material properties and environmental conditions.